

Unit 3 Practice Problems

Triangles - Properties and Congruence Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 3 Practice Problems

Directions

Work all problems first. Then check the answer bank. Detailed solutions are included after the answer bank.

Problems

1. A triangle has angle measures $x + 25$, $2x + 15$, and $3x - 10$. Find x and all three angle measures.
2. A triangle has remote interior angles 48° and 67° . Find the exterior angle.
3. An exterior angle measures $6x + 5$. The remote interior angles measure $2x + 30$ and $3x - 10$. Find x and the exterior angle.
4. In an isosceles triangle, the base angles measure $4x + 7$ and $2x + 31$. Find x , the base angles, and the vertex angle.
5. Two sides of a triangle are 7 and 9. Write the range of possible values for the third side x .
6. Classify by sides the triangle with $A(0, 0)$, $B(6, 0)$, and $C(3, 4)$.
7. Determine whether $A(0, 0)$, $B(8, 0)$, and $C(0, 6)$ form a right triangle.
8. If $AB \cong DE$, $BC \cong EF$, and $AC \cong DF$, what congruence theorem proves $\triangle ABC \cong \triangle DEF$?
9. If $AB \cong DE$, $AC \cong DF$, and $\angle A \cong \angle D$, what congruence theorem applies?
10. Does SSA prove triangle congruence in general?
11. If two angles and the included side are congruent, which triangle congruence theorem applies?
12. What information is needed to use HL?
13. If $\triangle ABC \cong \triangle DEF$, which angle corresponds to $\angle C$?
14. Given $AB \cong AD$, $CB \cong CD$, and shared side AC , prove $\triangle ABC \cong \triangle ADC$ and state one CPCTC conclusion.
15. A segment joins the midpoints of two sides of a triangle. If the third side is 18, find the midsegment length and its relationship to the third side.
16. A triangle midsegment is $3x - 2$, and the parallel third side is 20. Find x .
17. A median of a triangle has length 18. How far is the centroid from the vertex and from the midpoint of the opposite side?
18. For $A(2, 4)$, $B(8, 10)$, $C(10, 0)$, find the midpoints of AB and AC . Then verify the segment connecting those midpoints is parallel to BC and half its length.
19. A triangle has two angles 36° and 74° . Find the third angle.
20. Two lines intersect inside a proof, creating a pair of vertical angles. What reason proves the vertical angles congruent?
21. Use distance to classify $\triangle ABC$ with $A(-2, 1)$, $B(4, 1)$, $C(1, 5)$ by sides.

- 22.** Classify a triangle with side lengths 7, 24, 25 by angles.
- 23.** In $\triangle ABC$, $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. Write the equation of the altitude from C to $\overline{AB}$.
- 24.** What does it mean to prove triangle congruence using rigid motions?
- 25.** State the isosceles base angles theorem and its converse.

Answer Bank

1. $x = 25$; angles $50^\circ, 65^\circ, 65^\circ$.
2. 115° .
3. $x = 15$; exterior angle 95° .
4. $x = 12$; base angles 55° , vertex angle 70° .
5. $2 < x < 16$.
6. Isosceles.
7. Yes, right triangle.
8. SSS.
9. SAS.
10. No.
11. ASA.
12. Right triangles, congruent hypotenuses, and one pair of congruent legs.
13. $\angle F$.
14. SSS; for example $\angle BAC \cong \angle CAD$.
15. Length 9; parallel to the third side.
16. $x = 4$.
17. Vertex to centroid 12, centroid to midpoint 6.
18. Midpoints $(5, 7)$ and $(6, 2)$; slopes both -5 , lengths $\sqrt{26}$ and $2\sqrt{26}$.
19. 70° .
20. Vertical Angles Theorem.
21. Isosceles.
22. Right triangle.
23. $x = 2$.
24. A sequence of translations, rotations, and reflections maps one triangle exactly onto the other.
25. If two sides are congruent, opposite angles are congruent; conversely, if two angles are congruent, opposite sides are congruent.

Detailed Solutions

1. Use the Triangle Sum Theorem: $(x + 25) + (2x + 15) + (3x - 10) = 180$. Combine like terms: $6x + 30 = 180$, so $6x = 150$ and $x = 25$. Substitute: 50° , 65° , and 65° .
2. By the Exterior Angle Theorem, the exterior angle equals the sum of the two remote interior angles: $48 + 67 = 115^\circ$.
3. Set the exterior angle equal to the sum of the remote interior angles: $6x + 5 = (2x + 30) + (3x - 10)$. Then $6x + 5 = 5x + 20$, so $x = 15$. The exterior angle is $6(15) + 5 = 95^\circ$.
4. Base angles in an isosceles triangle are congruent, so $4x + 7 = 2x + 31$. Then $2x = 24$, so $x = 12$. Each base angle is 55° . The vertex angle is $180 - 55 - 55 = 70^\circ$.
5. Use the triangle inequality shortcut: $|9 - 7| < x < 9 + 7$. Therefore $2 < x < 16$.
6. Compute distances. $AB = 6$. $AC = \sqrt{3^2 + 4^2} = 5$. $BC = \sqrt{(-3)^2 + 4^2} = 5$. Since two sides are equal, the triangle is isosceles.
7. Distances are $AB = 8$, $AC = 6$, and $BC = \sqrt{8^2 + (-6)^2} = 10$. Since $6^2 + 8^2 = 10^2$, the triangle is right.
8. The information gives three pairs of corresponding sides congruent, so SSS proves the triangles congruent.
9. The angle $\angle A$ is included between sides AB and AC , matching $\angle D$ between DE and DF . Therefore SAS applies.
10. SSA does not prove congruence in general because the same two sides and a non-included angle can sometimes form two different triangles.
11. Two angles and the included side are exactly ASA.
12. HL is for right triangles only. You need both triangles to be right triangles, their hypotenuses congruent, and one pair of corresponding legs congruent.
13. The congruence order $ABC \cong DEF$ matches $A \leftrightarrow D$, $B \leftrightarrow E$, and $C \leftrightarrow F$. Thus $\angle C$ corresponds to $\angle F$.
14. Use $AB \cong AD$, $CB \cong CD$, and $AC \cong AC$ by the Reflexive Property. Thus $\triangle ABC \cong \triangle ADC$ by SSS. Then corresponding parts are congruent by CPCTC, such as $\angle BAC \cong \angle CAD$.
15. The segment joining midpoints is a midsegment. A midsegment is parallel to the third side and half its length, so the length is $18/2 = 9$.
16. The midsegment is half the third side, so $3x - 2 = 10$. Then $3x = 12$, so $x = 4$.
17. The centroid divides each median in a $2 : 1$ ratio from the vertex. For total median length 18, the vertex-to-centroid distance is $\frac{2}{3}(18) = 12$, and centroid-to-midpoint distance is $\frac{1}{3}(18) = 6$.
18. Midpoint of AB is $\left(\frac{2+8}{2}, \frac{4+10}{2}\right) = (5, 7)$. Midpoint of AC is $(6, 2)$. Slope between midpoints is $(2 - 7)/(6 - 5) = -5$. Slope of BC is $(0 - 10)/(10 - 8) = -5$, so they are parallel. The midpoint segment length is $\sqrt{1^2 + (-5)^2} = \sqrt{26}$, while $BC = \sqrt{2^2 + (-10)^2} = \sqrt{104} = 2\sqrt{26}$.
19. The angles of a triangle add to 180° . The third angle is $180 - 36 - 74 = 70^\circ$.
20. Vertical angles are congruent by the Vertical Angles Theorem.
21. Distances: $AB = 6$, $AC = \sqrt{3^2 + 4^2} = 5$, and $BC = \sqrt{(-3)^2 + 4^2} = 5$. Two sides are congruent, so the triangle is isosceles.

22. Check the Pythagorean relationship with largest side 25: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Therefore it is a right triangle.
23. Segment AB is horizontal because both endpoints have $y = 0$. A line perpendicular to a horizontal line is vertical. Through $C(2, 4)$, the altitude is $x = 2$.
24. Rigid-motion congruence means one triangle can be moved by translations, rotations, and reflections without changing size or shape until it lands exactly on the other triangle.
25. The theorem says congruent sides in a triangle have congruent opposite angles. The converse says congruent angles in a triangle have congruent opposite sides.